

Geometric State Judgment vs. the per-turn LLM (VDM)

A short report: what we tested, how, and what we found

cap-x-se / track-judge experiment

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1. What we tested

An agentic robot “code-as-policies” loop writes code, runs it, then must *judge the environment state* each turn to decide **FINISH** (task done) or **REGENERATE** (revise the code). cap-x does this with a **VDM** — a Visual Differencing Model, i.e. a *second LLM call* every turn that looks at before/after images and returns a verdict.

Our question: can we *remove that per-turn LLM call* and judge state with **geometry instead** — code that the agent writes once, driven by local models (SAM segmentation + TAPIP3D 3D point tracking), with **zero LLM at judgment time**? And is it as reliable, while being cheaper (fewer LLM calls)?

2. How we tested it

The judge is one declarative line. The agent names a *relation* over *objects* in plain words; all perception/tracking/geometry is in deterministic tooling:

```
J.judge(ctx, "place_in", target="bowl", reference="plate")
```

A small set of relations (`place_in` / `on_top_of` / `opened` / ... + `all_of` for multi-step) covers **28 of 30** dev tasks.

We could not start the runtime LLM endpoint (a parallel migration occupies it), so for the *loop* experiments we **stood in for the runtime LLM ourselves** (wrote the one-line judge; stubbed execution by driving the simulator), and ran the *real* judgment harness. Three measurements:

- **GT-validated loops.** Run a task across turns; compare the geometric verdict to LIBERO’s ground-truth `task_completed()` each turn. Tasks: open a drawer (`opened`), put a bowl on a plate (`on_top_of`), and the compositional “open the top drawer *and* put the bowl inside” (`all_of`).
- **Accuracy sweep.** Sample many states by teleporting the bowl (seated-on-plate, clearly-off, hovering-too-high) across 3 seeds $\times$ 7 placements; record the judge verdict vs. ground truth $\rightarrow$ a confusion matrix.

- **Baseline cost.** Mine the official cap-x VDM run logs (N=68) for success rate, how often the agent never finishes, and the number of regenerations (each = one extra VDM call).

3. What we found

(a) **The geometric judge is reliable.** On the accuracy sweep it matched ground truth on **every** sampled state, with no false positives or false negatives — including the tricky “bowl hovering over the plate” case (a 2D frame-diff can mistake overlap for “placed”).

Geometric judge vs. ground truth (N = 21)	value
accuracy (verdict = ground truth)	21/21 = 100%
false “done” / missed “done”	0 / 0
precision / recall	1.00 / 1.00

(b) **The VDM baseline is poor — and the failure is in the *judgment*.**

Official cap-x VDM baseline (N = 68 runs)	value
ground-truth task success	12% (8/68)
runs that <i>never</i> FINISH (VDM never says “done”)	46%
successes the VDM missed (missed-done)	1 of 8
valid code written, task still fails	74%
mean regenerations per run (= extra VDM calls)	4.5

Nearly half the runs *never converge a verdict at all*: the VDM keeps saying “not done” and the agent regenerates until it runs out of turns. That is a judgment failure, not only a hard-to-execute task.

(c) **Fewer LLM calls.** The geometric judge makes **0 LLM calls per turn** vs. the VDM’s 1. With code generation at 1 call/turn, total LLM calls **halve**; and the baseline’s 4.5 regenerations/run are 4.5 extra VDM calls the geometric arm removes. The verdict is also **deterministic** (same state → same answer), unlike an LLM sampled at temperature.

(d) **It generalizes to a multi-step task cap-x fails.** On “open the top drawer and put the bowl inside” (which the VDM baseline fails on both perturbed suites), the one-line `all_of` judge tracked *both* subgoals and agreed with ground truth on every turn. Its per-turn message is a *metric residual* — e.g. “drawer open ✓, bowl still 0% inside” — which tells the next code revision exactly what remains. A frame-diff VDM cannot produce that, which is why multi-step tasks are where it fails worst.

4. Honest limits

The 100% accuracy is on a *modest, controlled* sample (teleported states, one relation, one task) — it measures the judge’s accuracy ceiling, not a same-loop head-to-head against the VDM (which needs the runtime LLM endpoint). The baseline numbers are one trial per task and noisy. The recurring weak spot is *not* the geometry but *grounding* — naming an object to the segmenter (small/branded items, or a drawer once it is open and occluded); we mitigate it with a confidence gate and a cross-turn cache, and a stronger grounder is the next lever. Both measurements are real and point the same way; the same-loop success+speed A/B is the remaining confirmation.